EXPERIMENT – 1

Analysis, Development and Verification of 8-bit and 16-bit Arithmetic Operations Using Assembly Language Program

Objective

To develop and execute 8-bit and 16-bit arithmetic operations using an Intel 8-bit microprocessor assembly language program and verify the obtained results.

Algorithm

1. Initialize the required registers and memory locations.
2. Load the input data into the appropriate registers.
3. Perform 8-bit addition and subtraction.
4. Perform 16-bit addition and subtraction using register pairs.
5. Perform multiplication using repeated addition.
6. Perform division using repeated subtraction.
7. Store the results in the specified memory locations.
8. Verify the results with manual hexadecimal calculations.

Software and Hardware Required

- 8085/8-bit Microprocessor Simulator
- 8-bit Microprocessor Trainer Kit
- Keyboard
- Power Supply

A) 8-bit Addition

Code

```
LDA 4500H
MOV B,A
LDA 4501H
ADD B
STA 4502H
MVI A,00H
ACI 00H
```

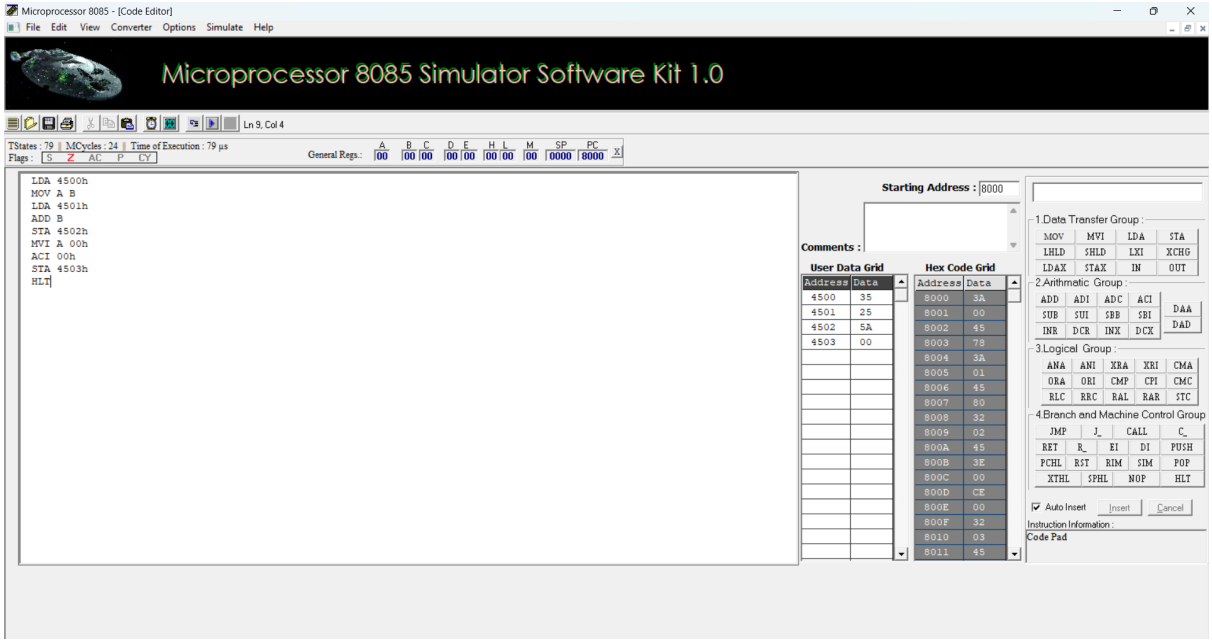
STA 4503H

HLT

Input

Address	Data
4500H	25H
4501H	35H

Output



Address	Data
4502H	5AH
4503H	00H

Calculation:

25H + 35H = **5AH**

B) 8-bit Subtraction

Code

LDA 4500H

MOV B,A

LDA 4501H

SUB B

STA 4502H

MVI A,00H

ACI 00H

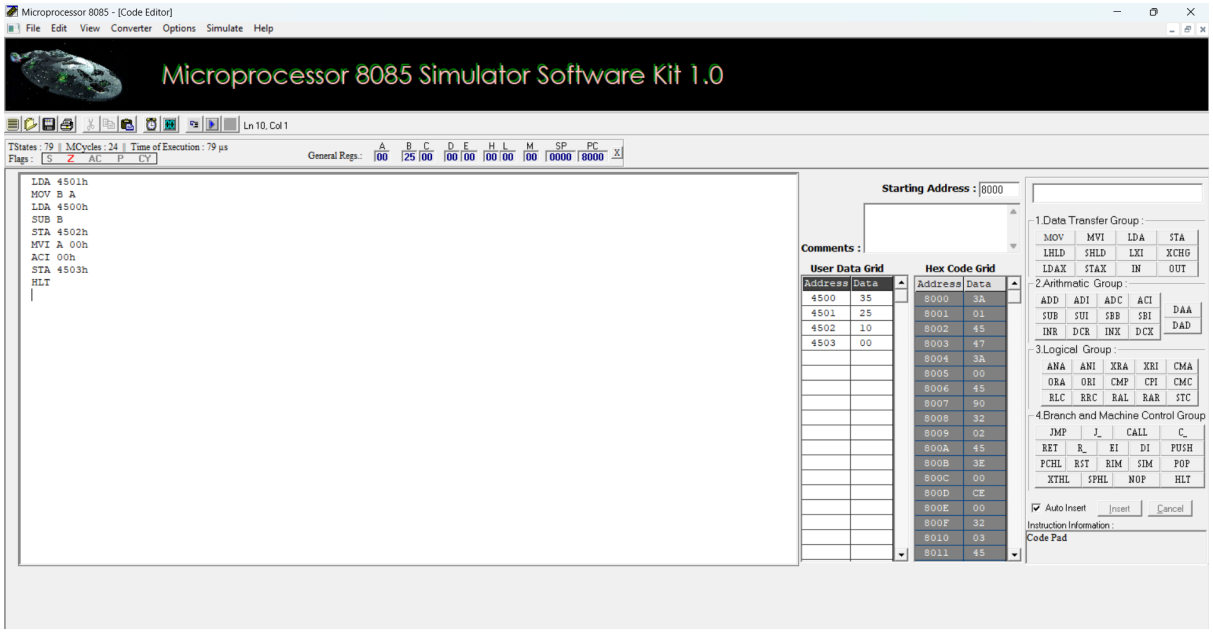
STA 4503H

HLT

Input

Address	Data
4500H	35H
4501H	25H

Output



Address	Data
4502H	10H
4503H	00H

Calculation:
35H − 25H = **10H**

C) 16-bit Multiplication

Code

LXI H,0012H

MOV D,H

MOV E,L

LXI H,0003H

MOV B,H

MOV C,L

LXI H,0000H

LOOP: DAD D

DCX B

MOV A,B

ORA C

JNZ LOOP

SHLD 4502H

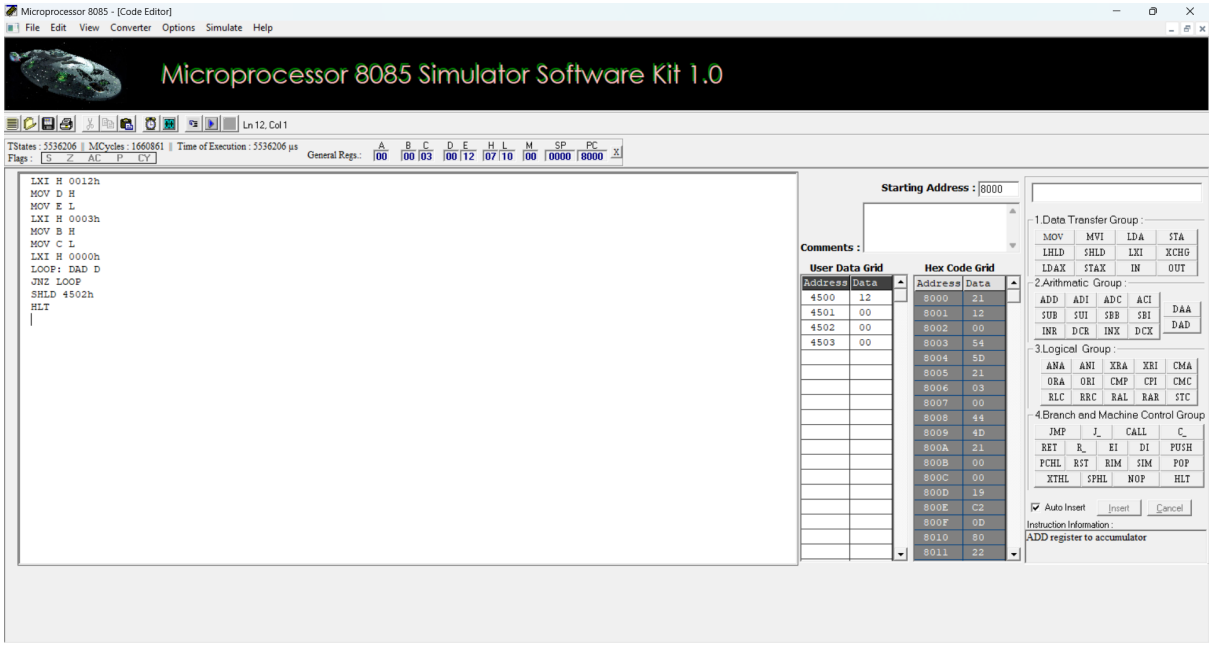
HLT

Input

Address	Data
4500H	12H
4501H	00H

Multiplier = **03H**

Output



Address	Data
4502H	36H
4503H	00H

Calculation:
12H × 03H = **36H**

D) 16-bit Division

Code

```
LXI H,0012H
MOV D,H
MOV E,L

LXI H,0003H
MOV B,H
MOV C,L

LXI H,0000H

LOOP: MOV A,L
SUB E
MOV L,A
MOV A,H
SBB D
MOV H,A
JC DONE
INX B
JMP LOOP

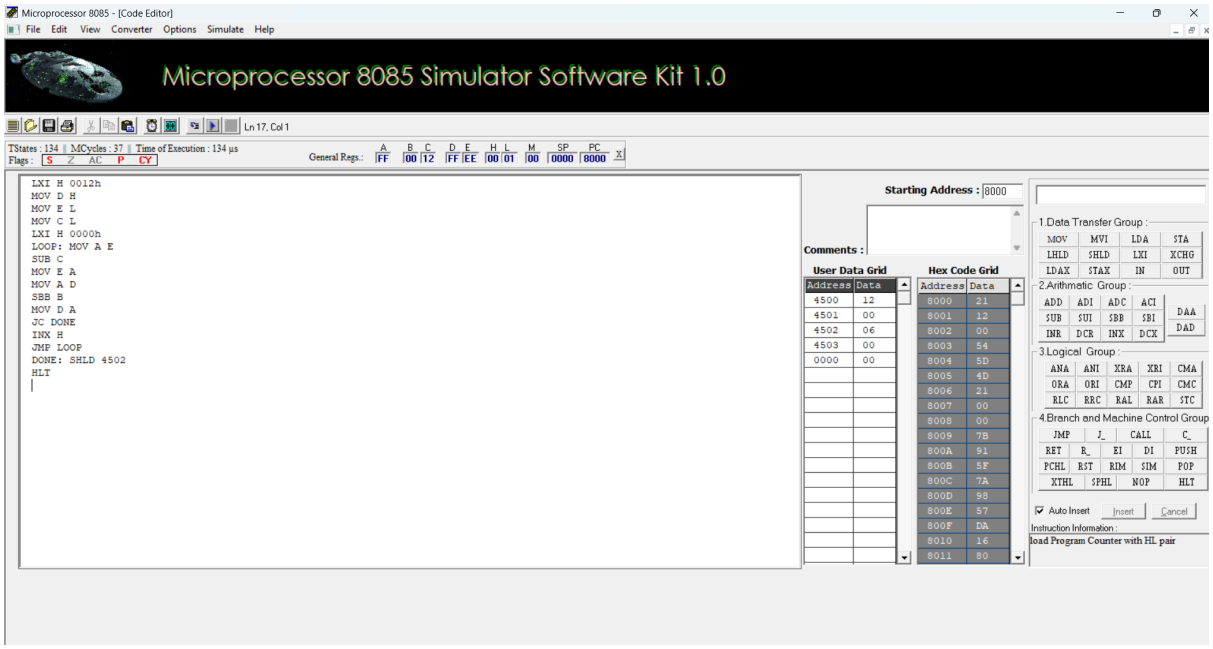
DONE: SHLD 4502H
```

HLT

Input

Address	Data
Dividend	12H
Divisor	03H

Output



Address	Data
4502H	06H
4503H	00H

Calculation:

12H ÷ 03H = **06H remainder 00H**

Result

Thus, the **8-bit and 16-bit arithmetic operations were successfully developed and verified using assembly language programming**. The obtained results were stored in the specified memory locations and verified using hexadecimal calculations.

Student Reflection

I learned how to perform arithmetic operations using 8-bit microprocessor assembly language. I understood the use of registers, register pairs, memory locations, carry and borrow during arithmetic operations. I also learned multiplication using repeated addition and division using repeated subtraction.